SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I R.Hashini / 1924260352 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

Answer to Question 1

Subnet of 192.168.10.65/26: Blocks of size 64 start at .0, .64, .128 and .192. The address .65 falls in 192.168.10.64/26.

Network address: 192.168.10.64 **Broadcast address:** 192.168.10.127 **Valid host range:** 192.168.10.65 to 192.168.10.126 (62 hosts)

Validity check: The workstation IP .65 is a valid host address in its own subnet. The gateway 192.168.10.1 is not: it belongs to 192.168.10.0/26 (hosts .1 to .62), so it lies outside the workstation subnet and cannot be reached directly.

Configuration errors: (1) The gateway is in a different subnet from the workstation. (2) The workstation sits in 192.168.10.64/26, while the systems it must reach are in 192.168.10.0/26, so the two hosts are on different subnets and cannot talk without a correct gateway.

Corrected configuration: IP address 192.168.10.10 with subnet mask 255.255.255.192 (/26) and default gateway 192.168.10.1. The workstation is now in 192.168.10.0/26 with the other systems and uses the correct gateway.

Usable host addresses after correction: 62 (192.168.10.1 to 192.168.10.62). One of these is the gateway, so 61 addresses remain for end hosts.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer to Question 2

Expanded Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329/64

Expanded Device B: 2001:0db8:0000:0000:0001:0000:0000:0001/64

Network prefixes (first 64 bits): Device A is in 2001:0db8:0000:0000::/64. Device B is in 2001:0db8:0000:0000::/64.

Same subnet? Yes. Both addresses share the same /64 prefix, so as written they are on the same subnet.

Prefix mismatch: No mismatch is visible in these addresses. The "1" in the fifth group of Device B falls in the interface-ID half of the address, not the subnet part. If the intended subnet for Device B was 2001:db8:0:1::/64, the address was typed incorrectly and should be 2001:db8:0:1::1/64. Otherwise, check for other causes: IPv6 not enabled on the interface, a disabled interface, a duplicate address, or ICMPv6 blocked by a firewall.

Optimized configuration: Device A: 2001:db8::ff00:42:8329/64 (unchanged). Device B: 2001:db8::2/64, which expands to 2001:0db8:0000:0000:0000:0000:0000:0002/64. Both are in the same subnet, 2001:db8::/64.

Available host addresses in the /64 subnet: 2^64 addresses (18,446,744,073,709,551,616). The all-zero interface ID is reserved as the subnet-router anycast address, leaving 2^64 - 1 = 18,446,744,073,709,551,615 assignable addresses.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

Answer to Question 3

Assumption: The question does not list the departmental host counts, so the figures below are assumed for illustration: Engineering 100, Sales 50, Finance 25, HR 12, Admin 6, and one WAN link needing 2 hosts. The method applies to any list of requirements.

Current (inefficient) allocation

Department	Subnet	Network	Broadcast	Host range	Usable	Needed	Unused / Shortfall
Engineering	/26	192.168.1.0	192.168.1.63	.1 to .62	62	100	Short by 38
Sales	/26	192.168.1.64	192.168.1.127	.65 to .126	62	50	12 unused
Finance	/27	192.168.1.128	192.168.1.159	.129 to .158	30	25	5 unused
HR	/27	192.168.1.160	192.168.1.191	.161 to .190	30	12	18 unused
Admin	/28	192.168.1.192	192.168.1.207	.193 to .206	14	6	8 unused
WAN link	/28	192.168.1.208	192.168.1.223	.209 to .222	14	2	12 unused
Unallocated	/27	192.168.1.224	192.168.1.255	.225 to .254	30	0	30 unused

Inefficiencies and errors: Engineering is given a /26 (62 hosts) but needs 100, so it needs a /25. The WAN link uses a /28 with 14 hosts for only 2 needed, and HR uses a /27 for 12 people. A /27 block (30 addresses) is left unallocated. Blocks were assigned in the wrong order, so large requirements were not placed first.

Usable and unused totals: The current plan has 212 usable addresses assigned to departments against 195 needed, so 17 are unused within assigned blocks, and 30 more are unallocated. The main problem is that Engineering is short while other blocks are oversized.

Optimized VLSM allocation: Allocate the largest requirement first and size each block to its need.

Department	Need	Subnet	Network	Broadcast	Host range	Usable	Unused
Engineering	100	/25	192.168.1.0	192.168.1.127	.1 to .126	126	26
Sales	50	/26	192.168.1.128	192.168.1.191	.129 to .190	62	12
Finance	25	/27	192.168.1.192	192.168.1.223	.193 to .222	30	5
HR	12	/28	192.168.1.224	192.168.1.239	.225 to .238	14	2
Admin	6	/29	192.168.1.240	192.168.1.247	.241 to .246	6	0
WAN link	2	/30	192.168.1.248	192.168.1.251	.249 to .250	2	0
Spare	0	/30	192.168.1.252	192.168.1.255	.253 to .254	2	2 reserved

Result: All six departments fit. The 192.168.1.0/24 network is fully used with 240 usable addresses assigned to departments (195 needed) and 2 kept spare for growth. Each block is aligned on its own boundary, so the plan can be summarized later.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer to Question 4

Question 4 uses the same addresses as Question 2, so the answer is the same.

Expanded addresses: Device A 2001:0db8:0000:0000:0000:ff00:0042:8329/64; Device B 2001:0db8:0000:0000:0001:0000:0000:0001/64.

Prefix and subnet: Both are in 2001:0db8:0000:0000::/64, so they are on the same subnet as written. No prefix mismatch is present; check the interface state, IPv6 enablement and firewall rules, and confirm the intended subnet for Device B.

Optimized addresses: Device A 2001:db8::ff00:42:8329/64; Device B 2001:db8::2/64.

Available hosts: 2^64 addresses per /64, of which 2^64 - 1 = 18,446,744,073,709,551,615 are assignable.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

Answer to Question 5

Matching entries for 192.168.2.100: 192.168.2.0/24 matches (the first 24 bits match) and the default route 0.0.0.0/0 also matches with 0 bits. 192.168.1.0/24 does not match.

Longest-prefix match: 192.168.2.0/24 is selected because /24 is longer than the default route.

Next hop: 10.0.0.3

Intended subnet check: 192.168.2.100 is inside 192.168.2.0 to 192.168.2.255, so the destination belongs to the intended subnet and the /24 entry is correct.

Routing errors: (1) The table is correct on paper, so the failure is most likely at 10.0.0.3: check that it is reachable from the router (ping 10.0.0.3), that it has a route to 192.168.2.0/24, and that

the host 192.168.2.100 is up. (2) The default route is written as "Default Route" rather than 0.0.0.0/0, which is not valid notation. (3) If the 192.168.2.0/24 route is ever missing or its next hop fails, traffic silently falls back to the default route via 10.0.0.1, which hides the fault.

Optimization: Write the default route as 0.0.0.0/0 via 10.0.0.1 and add an exit interface or verify each next hop is reachable. 192.168.1.0/24 and 192.168.2.0/24 cannot be summarized into one route because they use different next hops and do not form an aligned supernet on their own.

Route used when no destination matches: The default route 0.0.0.0/0 via 10.0.0.1.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation